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Expt - 8

Title - Dijkstra's Algorithm

```
In [1]: 1 n = int(input("Enter number of vertices in the graph: "))
2
3 print("Enter the adjacency matrix where each row separated by spaces:")
4 graph = []
5 for i in range(n):
6     row = input().split()
7     graph.append([int(x) for x in row])
8
9 source = int(input("Enter the source vertex: "))
10
11 def dijkstra(graph, source):
12     V = len(graph)
13
14     dist = [float('inf')] * V
15     dist[source] = 0
16
17     visited = [False] * V
18
19     for _ in range(V):
20         min_dist = float('inf')
21         min_index = -1
22
23         for v in range(V):
24             if not visited[v] and dist[v] < min_dist:
25                 min_dist = dist[v]
26                 min_index = v
27
28         visited[min_index] = True
29
30         for v in range(V):
31             if not visited[v] and graph[min_index][v]:
32                 dist[v] = min(dist[v], dist[min_index] + graph[min_index][v])
33
34     return dist
35
36 Shortest_Paths = dijkstra(graph, source)
37 print("=====")
38 print("Shortest distances from node", source, "to each node:")
39
40 for i, j in enumerate(Shortest_Paths):
41     print("Vertex", i, "Distance =", j)
42 print("=====")
```

```
Enter number of vertices in the graph: 5
Enter the adjacency matrix where each row separated by spaces:
0 0 0 7 0
3 0 4 0 0
0 0 0 5 6
0 2 0 0 0
0 0 0 4 0
Enter the source vertex: 0
=====
Shortest distances from node 0 to each node:
Vertex 0 Distance = 0
Vertex 1 Distance = 9
Vertex 2 Distance = 13
Vertex 3 Distance = 7
Vertex 4 Distance = 19
=====
```